Minimum-Thrust Planar Rocket Trajectory Using Euler–Lagrange Equations

Amir Khan

Abstract

This project investigates a simplified planar rocket trajectory problem using the Euler–Lagrange equation. A quadratic thrust-effort (u(t)) cost functional is minimised to determine the trajectory between two fixed points. The resulting optimal trajectory, thrust profile, and flight time are derived analytically and interpreted physically.

Determine a trajectory $(x(t), y(t))$ that minimises total thrust effort.

1 Introduction

The objective of this project is to determine the optimal time required for a rocket to reach a specified final position while minimising the associated thrust cost.

This is formulated as an optimisation problem: find a trajectory $(x(t), y(t))$ that minimises the total thrust effort over the flight.

An Euler–Lagrange framework is used to solve this problem. By expressing the cost as a functional of the trajectory, the optimality conditions are obtained through the calculus of variations, leading to equations that characterise the optimal motion.

2 Model and Assumptions

- Planar motion.
- Rocket modelled as a point mass.
- Constant gravitational acceleration g.
- No aerodynamic drag.
- No mass variation.
- Thrust freely controllable.

3 Mathematical Model

$$\boldsymbol{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

$$\ddot{\boldsymbol{r}} = \boldsymbol{u}(t) - g\hat{\boldsymbol{y}}$$

$$\boldsymbol{u}(t) = \ddot{\boldsymbol{r}} + g\hat{\boldsymbol{y}}$$

4 Cost Functional

$$J = \int_0^T \|\boldsymbol{u}(t)\|^2 dt$$

$$J = \int_0^T \left[\ddot{x}^2 + (\ddot{y} + g)^2\right] dt$$

5 Euler–Lagrange Equations

State higher-order form:

$$\frac{\partial F}{\partial q} - \frac{d}{dt}\left(\frac{\partial F}{\partial \dot{q}}\right) + \frac{d^2}{dt^2}\left(\frac{\partial F}{\partial \ddot{q}}\right) = 0$$

Apply to both variables:

$$x^{(4)} = 0, \quad y^{(4)} = 0$$

6 General Solution

$$x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

$$y(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3$$

7 Boundary Conditions

$$x(0) = 0, \quad \dot{x}(0) = 0, \quad x(T) = L, \quad \dot{x}(T) = 0$$

$$y(0) = 0, \quad \dot{y}(0) = 0, \quad y(T) = H, \quad \dot{y}(T) = 0$$

Solution:

$$x(t) = L\left(3\frac{t^2}{T^2} - 2\frac{t^3}{T^3}\right)$$

$$y(t) = H\left(3\frac{t^2}{T^2} - 2\frac{t^3}{T^3}\right)$$

8 Thrust Profile

$$\ddot{x}(t) = \frac{6L}{T^2} - \frac{12L}{T^3}t$$

$$\ddot{y}(t) = \frac{6H}{T^2} - \frac{12H}{T^3}t$$

$$\boldsymbol{u}(t) = \begin{bmatrix} \frac{6L}{T^2} - \frac{12L}{T^3}t \\ \frac{6H}{T^2} - \frac{12H}{T^3}t + g \end{bmatrix}$$

9 Cost as a Function of Time

$$J(T) = \frac{12(L^2 + H^2)}{T^3} + g^2 T$$

10 Optimal Flight Time

$$\frac{dJ}{dT} = 0 \Rightarrow T^4 = \frac{36(L^2 + H^2)}{g^2}$$

$$T^* = \left(\frac{36(L^2 + H^2)}{g^2}\right)^{1/4}$$

11 Minimum Cost

$$J_{\min} = \frac{4\sqrt{6}}{3} g^{3/2} (L^2 + H^2)^{1/4}$$

12 Numerical Results and Plots

The analytical solution was evaluated using example values for L, H, and g. The resulting plots show the trajectory, position components, thrust components, and thrust magnitude.

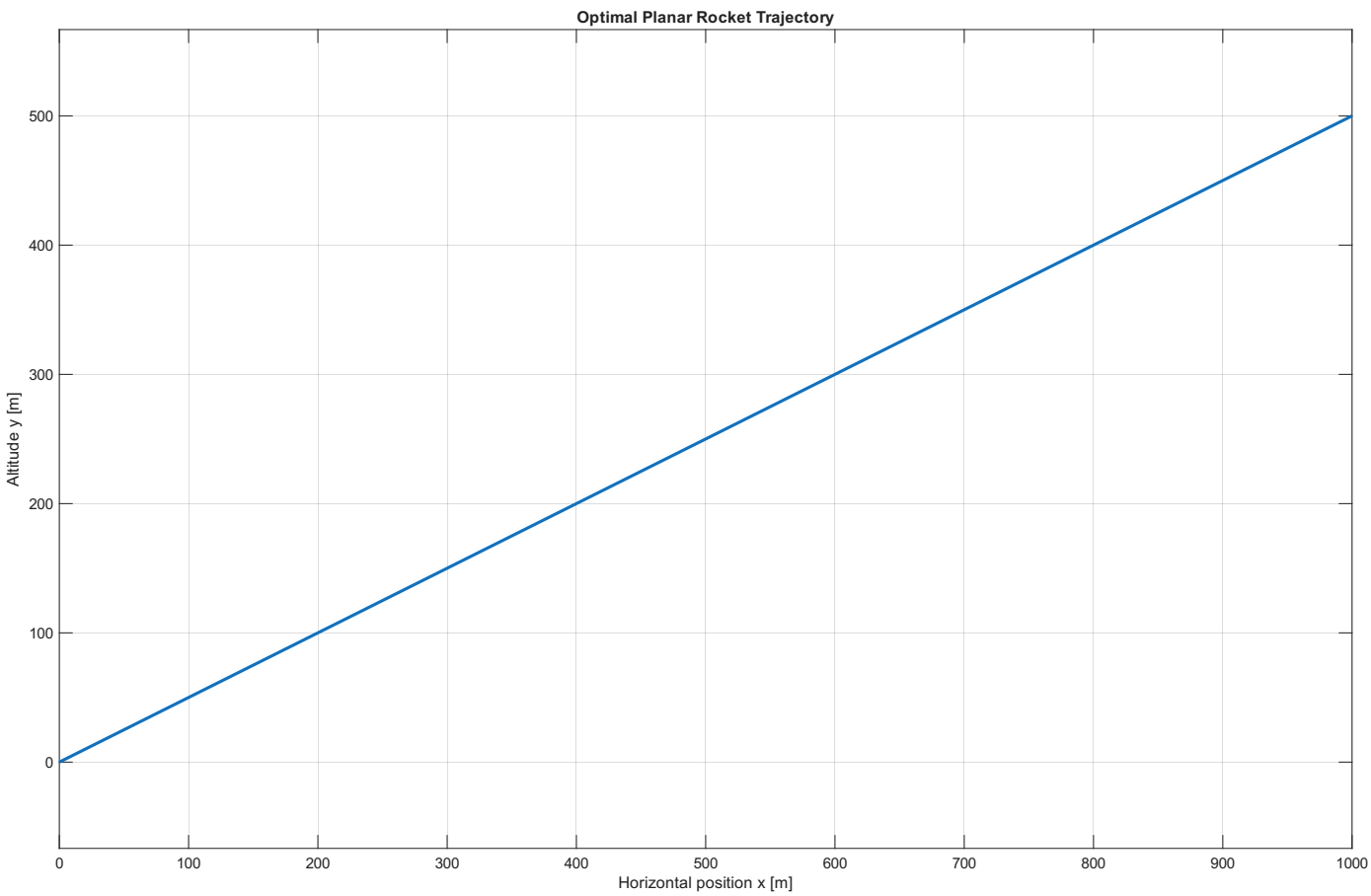

Figure 1: Optimal trajectory from $(0, 0)$ to (L, H).

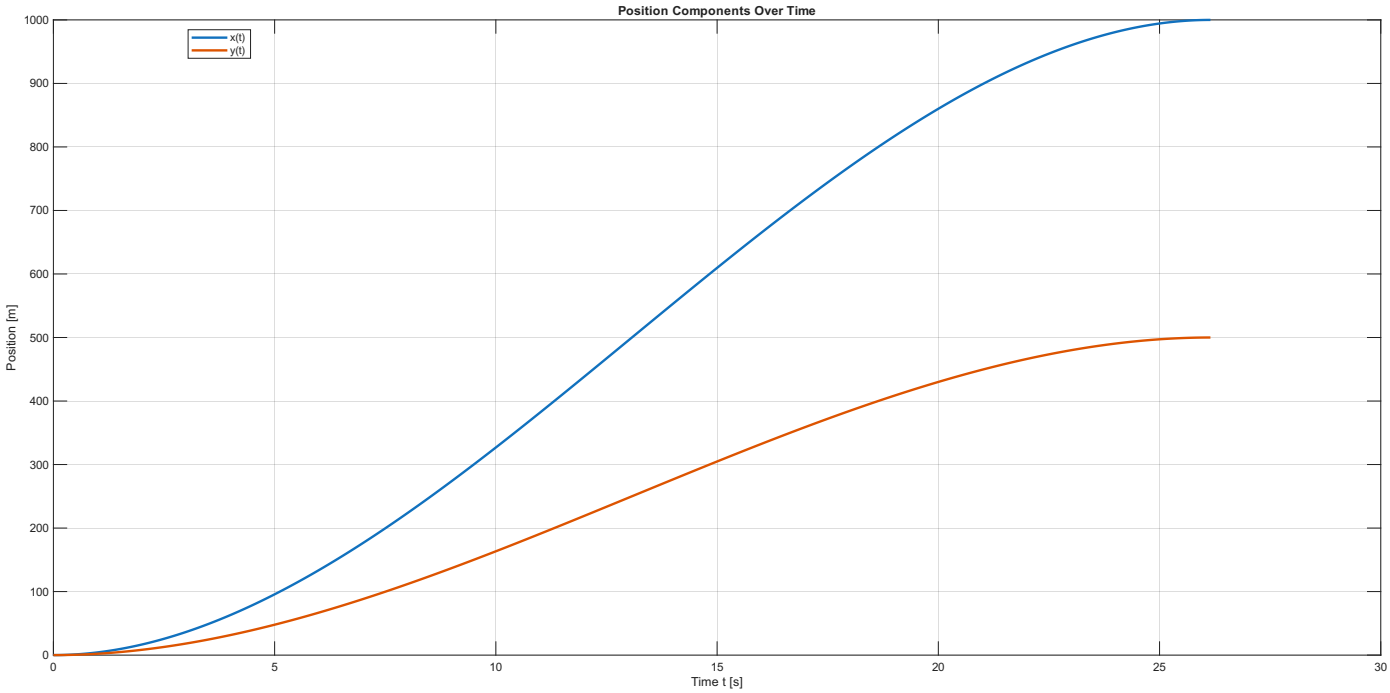

Figure 2: Horizontal and vertical position components over time.

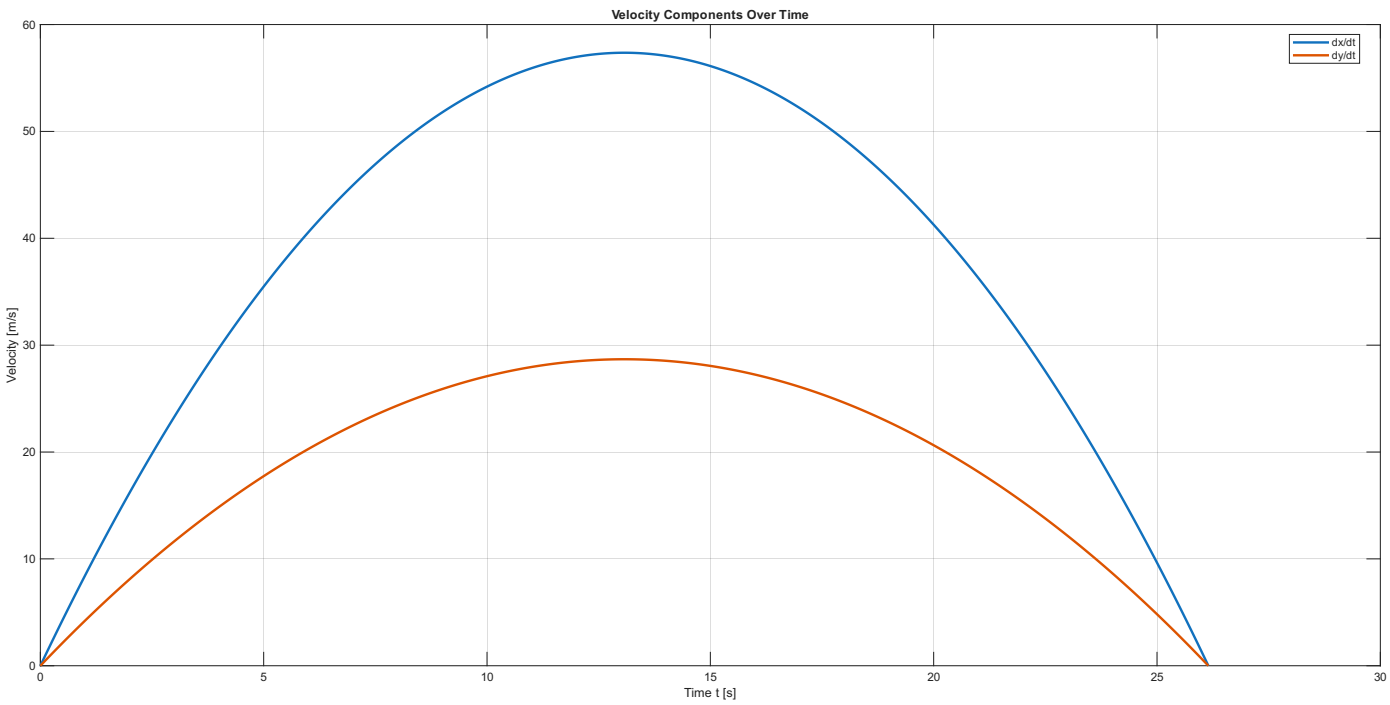

Figure 3: Velocity components over time.

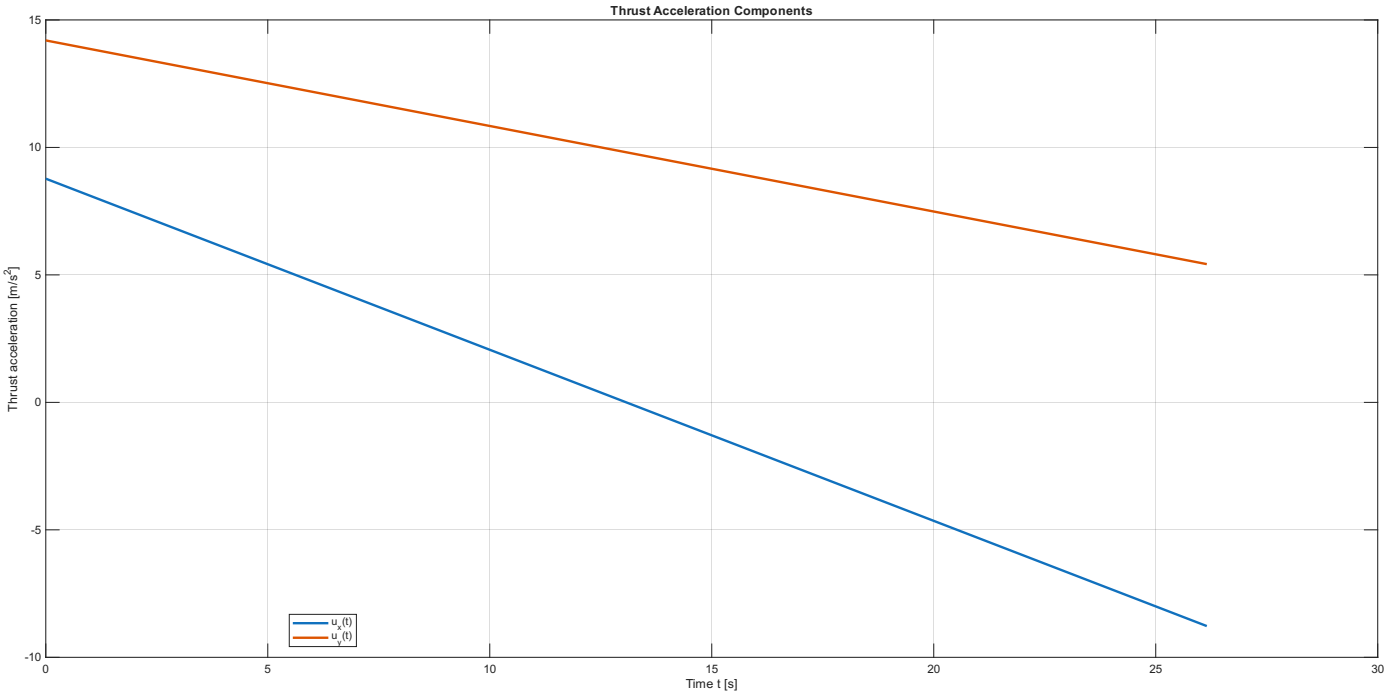

Figure 4: Required thrust acceleration components $u_x(t)$ and $u_y(t)$.

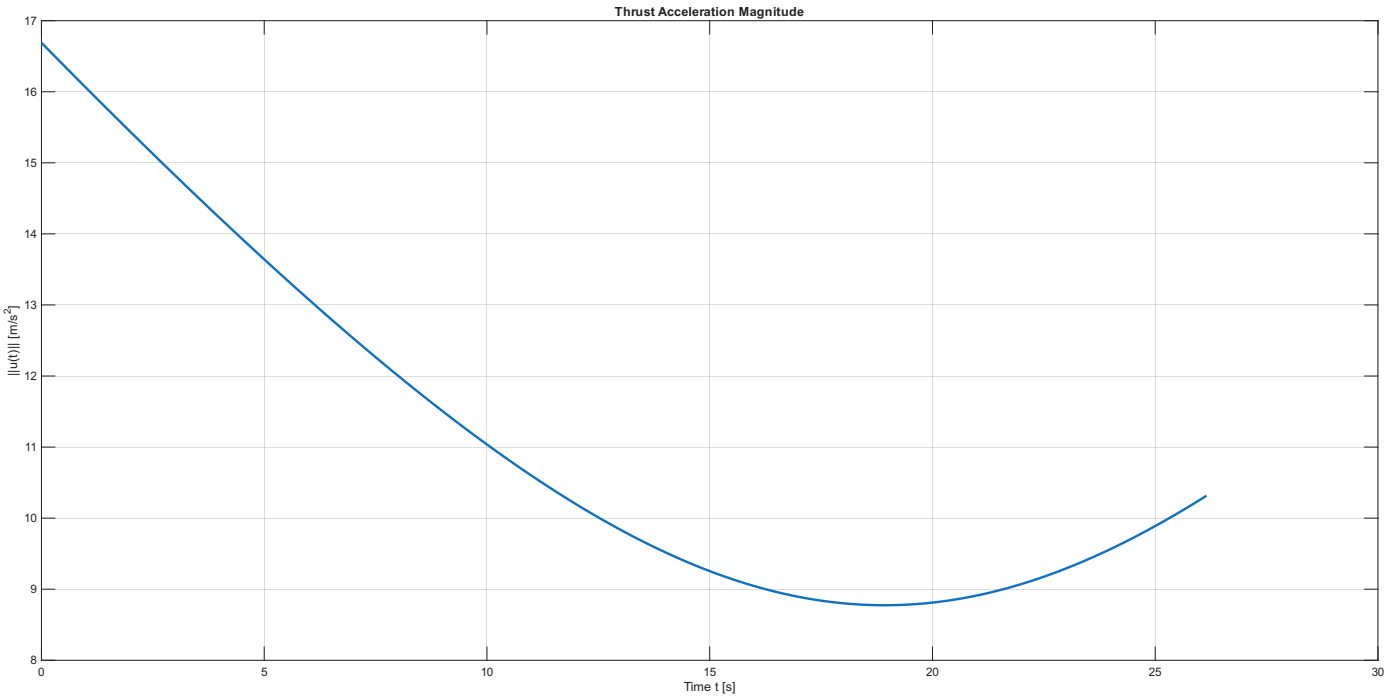

Figure 5: Magnitude of the thrust acceleration $\|\boldsymbol{u}(t)\|$.

13 Limitations

- No drag

- No mass change
- No thrust constraints
- Constant gravity
- 2D model only

14 Conclusion

In this project, a simplified planar rocket trajectory problem was analysed using the Euler–Lagrange framework. A quadratic thrust-effort cost functional was defined, reducing the problem to a variational optimisation over the trajectory functions $x(t)$ and $y(t)$.

Hence the resulting Euler–Lagrange equations led to fourth order differential equations, whose solutions were shown to be cubic polynomials in time. By applying appropriate boundary conditions, a unique trajectory connecting the initial and final states were retrieved.

The corresponding thrust profile was derived, allowing the total cost to be expressed as a function of the final time. This revealed a trade-off between fast abrupt motion, which requires large accelerations, and slower motion, which increases the cost of counteracting gravity force. Minimising this cost yielded an analytical expression for the optimal flight duration.

The results show that the optimal trajectory follows a straight-line path in space while exhibiting a smooth, time-dependent velocity profile. Although the model is simplified, it demonstrates how variational methods can be used to derive physically meaningful optimal trajectories.